Observability Metrics in Container Orchestration Environments for Microservices: A Systematic Mapping Study

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This study aims to identify and classify observability signals, tools, and usage contexts in container orchestration environments for microservices, examining how these signals support reliability, availability, resilience, fault diagnosis and recovery, and anomaly and attack detection.

Planning

Identify and classify observability signals, tools, and usage contexts in container orchestration environments for microservices, examining how these signals support reliability, availability, resilience, fault diagnosis and recovery, and anomaly and attack detection.

PICOC

Population: Microservices-based applications and systems

Intervention: Observability signals—metrics, logs, traces, and events—and collection and analysis mechanisms

Comparison: Comparison of different observability tools (I am not sure whether such a comparison is feasible)

Outcome: Uses related to reliability, availability, resilience, fault diagnosis and recovery, security, and anomaly or attack detection

Context: Containerized and orchestrated microservice environments on any relevant platform

Research Questions

RQ1. Which observability signals and metrics are used in container orchestration environments?

RQ2. Which platforms and tools are used, and what roles do they play in collection, storage, visualization, and analysis?

RQ3. In which contexts, and how, do these signals and metrics support reliability, availability, resilience, and fault detection, diagnosis, and recovery?

RQ4. What are the main challenges or gaps associated with implementing observability metrics in container orchestration environments?

Keywords and Synonyms

Keyword	Synonyms

Search String

```
(observab* OR monitor* OR telemetr* )
AND
(metric* OR log* OR trac* OR event* )
AND
( microserv* OR "micro-serv*" OR "micro serv*" OR "cloud native" OR "cloud-native" )
AND
( container* AND ( orchestrat* OR cluster* ) )
```

Sources

ACM Digital Library (http://portal.acm.org)

IEEE Digital Library (http://ieeexplore.ieee.org)

ISI Web of Science (http://www.isiknowledge.com)

Scopus (http://www.scopus.com)

Springer Link (http://link.springer.com)

Selection Criteria

Inclusion Criteria

- Collect, expose, or analyze at least one observability signal
- Address microservices in a containerized or orchestrated environment
- Use that signal for monitoring, diagnosis, dependability, security, or operational management

Exclusion Criteria

- Articles published before 2021
- Insufficient information to verify eligibility
- Does not address microservices or does not involve containerized applications in an orchestration environment
- Does not address microservices or does not involve containerized applications in an orchestration environment. A generic distributed system is insufficient.
- Not a scientific article
- Duplicate record
- Unrelated to the study focus
- Full text unavailable

Quality Assessment Checklist

Questions

Q1. Are the study objective and contribution clearly presented?

Q2. Are the observability signals and metrics identified and defined, including their source and collection level, when reported?

Q3. Is the study context—application, microservices architecture, and orchestration environment—described in sufficient detail?

Q4. Is the method for analyzing, classifying, or correlating signals explained clearly?

Q5. Are the conclusions and main claims supported by results, data, or examples presented in the study?

Q6. Are the tools or methods for collecting and analyzing metrics clearly described (e.g., how they work or are applied)?

Q7. Are limitations or threats to validity discussed?

Q8. Does the study provide sufficient details or artifacts to verify or attempt to reproduce its evaluation, where applicable?

Answers

- Yes
- Partially
- No
- Not applicable

Data Extraction Form

- Authors
- Publication Date
- Study Objective
- Study Type
- Application Domain
- Orchestration System
- Does the Study Involve Microservices
- Tools and Technologies Involved
- Tool Functions
- Collection and Analysis Approach
- Reported Signals or Metrics
- Metric Application Context
- Observability Signal Type
- Signal Sources
- Monitored Level
- Purpose of Observability Use
- Analysis or Detection Technique
- Action Associated with the Signals
- Evaluation Method
- Workload Dataset or Test Environment
- Reported Overhead or Cost
- Limitations and Threats to Validity
- Main Results
- Contributions

Conducting the Study

Digital Library Search Strings

Imported Studies

ACM Digital Library: 1451

IEEE Digital Library: 205

ISI Web of Science: 70

Scopus: 128

Springer Link: 1003